



3

Thermal physics

3.1 Thermal concepts

Assessment statements

- 3.1.1 State that temperature determines the direction of thermal energy transfer between two objects.
- 3.1.2 State the relation between the Kelvin and Celsius scales of temperature.
- 3.1.3 State that the internal energy of a substance is the total potential energy and random kinetic energy of the molecules of the substance.
- 3.1.4 Explain and distinguish between the macroscopic concepts of temperature, internal energy and thermal energy (heat).
- 3.1.5 Define the mole and molar mass.
- 3.1.6 Define the Avogadro constant.

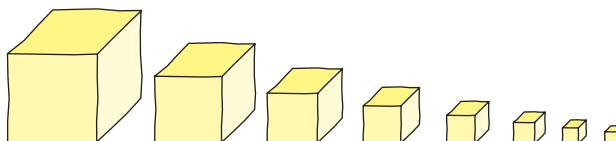
The role of the physicist is to observe our physical surroundings, take measurements and think of ways to explain what we see. Up to this point in the course we have been dealing with the motion of bodies. We can describe bodies in terms of their mass and volume, and if we know their speed and the forces that act on them, we can calculate where they will be at any given time. We even know what happens if two hit each other. However, this is not enough to describe all the differences between objects. For example, by simply holding different objects, we can feel that some are hot and some are cold.

In this chapter we will develop a model to explain these differences, but first of all we need to know what is inside matter.

The particle model of matter

Ancient Greek philosophers spent a lot of time thinking about what would happen if they took a piece of cheese and kept cutting it in half.

Figure 3.1 Can we keep cutting the cheese for ever?

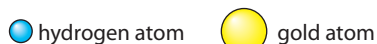


They didn't think it was possible to keep halving it for ever, so they suggested that there must exist a smallest part – this they called the *atom*.

Atoms are too small to see (about 10^{-10} m in diameter) but we can think of them as very small perfectly elastic balls. This means that when they collide, both momentum and kinetic energy are conserved.

Elements and compounds

We might ask: 'If everything is made of atoms, why isn't everything the same?' The answer is that there are many different types of atom.



There are 117 different types of atom, and a material made of just one type of atom is called an *element*. There are, however, many more than 117 different types of material. The other types of matter are made of atoms that have joined together to form molecules. Materials made from molecules that contain more than one type of atom are called *compounds*.

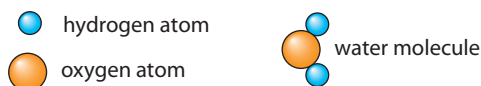


Figure 3.2 Gold is made of gold atoms and hydrogen is made of hydrogen atoms.

 This is a good example of how models are used in physics. Here we are modelling something that we can't see, the atom, using a familiar object, a rubber ball.

Figure 3.3 Water is an example of a compound.

The mole

When buying apples, you can ask for 5 kg of apples, or, say, 10 apples – both are a measure of amount. It's the same with matter – you can express amount in terms of either mass or number of particles.

A mole of any material contains 6.022×10^{23} atoms or molecules; this number is known as *Avogadro's number*.

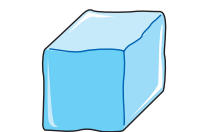
Although all moles have the same number of particles, they don't all have the same mass. A mole of carbon has a mass of 12 g and a mole of neon has a mass of 20 g – this is because a neon atom has more mass than a carbon atom.



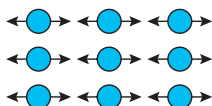
Moles of different compounds.

The three states of matter

From observations we know that there are three types, or *states* of matter: *solid*, *liquid* and *gas*. If the particle model is correct, then we can use it to explain why the three states are different.



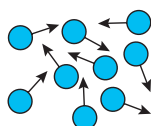
Solid Fixed shape and volume



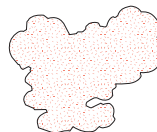
Molecules held in position by a force. Vibrate but don't move around.



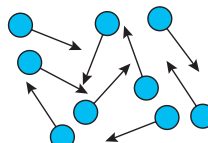
Liquid No fixed shape but fixed volume



Force between molecules not so strong so molecules can move around.



Gas No fixed shape or volume



No force between molecules (ideally).

Figure 3.4 The particle model explains the differences between solids, liquids and gases. (The arrows represent velocity vectors.)

 We can't prove that this model is true – we can only provide evidence that supports it.



Ice, water and steam.

● **Examiner's hint:** Be careful with the units. Do all calculations using m^3 .

Worked example

- 1 If a mole of carbon has a mass of 12 g, how many atoms of carbon are there in 2 g?
- 2 The density of iron is 7874 kg m^{-3} and the mass of a mole of iron is 55.85 g. What is the volume of 1 mole of iron?

Solution

- 1 One mole contains 6.022×10^{23} atoms.
2 g is $\frac{1}{6}$ of a mole so contains $\frac{1}{6} \times 6.022 \times 10^{23} \text{ atoms} = 1.004 \times 10^{23} \text{ atoms}$
- 2
$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$\text{volume} = \frac{\text{mass}}{\text{density}}$$

Volume of 1 mole = $\frac{0.05585}{7874} \text{ m}^3$
$$= 7.093 \times 10^{-6} \text{ m}^3$$

$$= 7.09 \text{ cm}^3$$

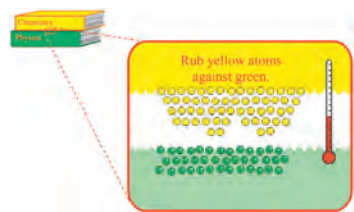
Exercises

- 1 The mass of 1 mole of copper is 63.54 g and its density 8920 kg m^{-3}
 - (a) What is the volume of one mole of copper?
 - (b) How many atoms does one mole of copper contain?
 - (c) How much volume does one atom of copper occupy?
- 2 If the density of aluminium is 2700 kg m^{-3} and the volume of 1 mole is 10 cm^3 , what is the mass of one mole of aluminium?

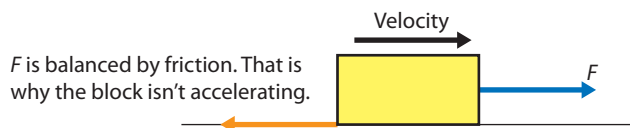
Internal energy

In Chapter 2, Mechanics, you met the concepts of energy and work. Use these concepts to consider the following:

Figure 3.5 A wooden block is pulled along a rough horizontal surface at a constant velocity.



To view a simulation showing how friction can increase temperature, visit heinemann.co.uk/hotlinks, enter the express code 4426P and click on Weblink 3.1.



Is any work being done on the block by force F ?

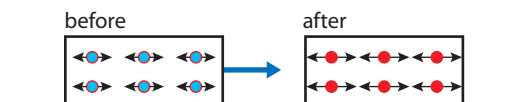
Is energy being transferred to the block?

Is the KE of the block increasing?

Is the PE of the block increasing?

Where is the energy going?

You will have realised that since work is done, energy is given to the block, but its PE and KE are not increasing. Since energy is conserved, the energy must be going somewhere. It is going inside the block as internal energy. We can explain what is happening using the particle model.

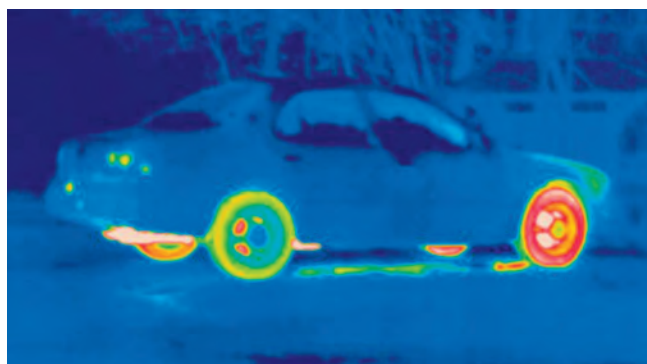


Molecules vibrate faster and are slightly further apart.

When we do work on an object, it enables the molecules to move faster (increasing KE) and move apart (increasing PE). We say that the *internal energy* of the object has increased.

Worked example

- 1 A car of mass 1000 kg is travelling at 30 m s^{-1} . If the brakes are applied, how much heat energy is transferred to the brakes?



Solution

When the car is moving it has kinetic energy. This must be transferred to the brakes when the car stops.

$$\begin{aligned} \text{KE} &= \frac{1}{2}mv^2 \\ &= \frac{1}{2} \times 1000 \times 30^2 \text{ J} \\ &= 450 \text{ kJ} \end{aligned}$$

So thermal energy transferred to the brakes = 450 kJ

Exercises

- 3 A block of metal, mass 10 kg, is dropped from a height of 40 m.
 (a) How much energy does the block have before it is dropped?
 (b) How much heat energy do the block and floor gain when it hits the floor?
- 4 If the car in Example 1 was travelling at 60 m s^{-1} , how much heat energy would the brakes receive?

Temperature

If we now pick a block up, after dragging it, we will notice something has changed. It has got hot; doing work on the block has made it hot. *Hotness* and *coldness* are the ways we perceive differences between objects. In physics, we use temperature to measure this difference more precisely.

Temperature (T) is a measure of how hot or cold an object is, and it is temperature that determines the direction of heat flow.

Temperature is a *scalar* quantity, and is measured in degrees Celsius ($^{\circ}\text{C}$) or kelvin (K).

Figure 3.6 Molecules gain internal energy.



In a solid, this means increasing the KE and PE of the molecules; in a gas it is just the KE. This is because there are no forces between the molecules of a gas, so it doesn't require any work to pull them apart.

This thermogram of a car shows how the wheels have become hot owing to friction between the road and the tyres, and the brakes pads and discs.

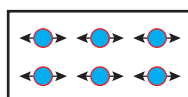


It is important to realise the difference between perception and physical measurement.

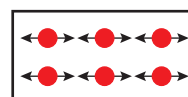
0 °C is equivalent to 273 K.
100 °C is equivalent to 373 K.

At normal atmospheric pressure,
pure water boils at 100 °C and
freezes at 0 °C. Room temperature
is about 20 °C.

Figure 3.7 Temperature is related to kinetic energy.



Cold – molecules
vibrate a bit.



Hot – molecules vibrate faster
and are slightly further apart.

During the first part of this chapter, we will measure temperature in Celsius. However when dealing with gases, we will use kelvin – this is because the Kelvin scale is based on the properties of a gas.

To convert from degrees Celsius to kelvin, simply add 273.

Thermometers

Temperature cannot be measured directly, so we have to find something that changes when the temperature changes. The most common thermometer consists of a small amount of alcohol in a thin glass tube. As temperature increases, the volume of the alcohol increases, so it rises up the tube. When we measure temperature, we are really measuring the length of the alcohol column, but the scale is calibrated to give the temperature in °C.

Temperature and the particle model

From the previous model, we can see that the particles in a hot body move faster than those in a cold one. The temperature is related to the average KE of the particles.

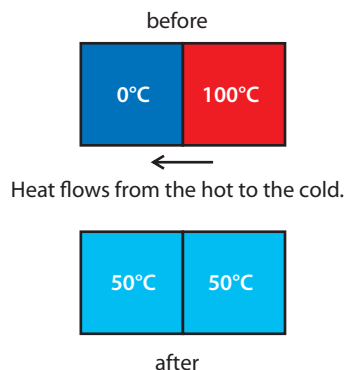
Heat transfer

Pulling a block of wood along a rough surface is not the only way to increase its temperature. We can make a cold body hot by placing it next to a hot body. We know that if the cold body gets hot, then it must have received energy – this is heat or *thermal* energy.

We are often more interested in preventing heat flow than causing it. Placing an insulating layer (e.g. woollen cloth) between the hot and cold bodies will reduce the rate of heat flow.

Thermal equilibrium

Figure 3.8 Heat flows from the hot body to the cold body until they are at the same temperature.



At this point no more heat will flow – this is called *thermal equilibrium*.

3.2 Thermal properties of matter

Assessment statements

- 3.2.1 Define *specific heat capacity* and *thermal capacity*.
- 3.2.2 Solve problems involving specific heat capacities and thermal capacities.
- 3.2.3 Explain the physical differences between the solid, liquid and gaseous phases in terms of molecular structure and particle motion.
- 3.2.4 Describe and explain the process of phase changes in terms of molecular behaviour.
- 3.2.5 Explain in terms of molecular behaviour why temperature does not change during a phase change.
- 3.2.6 Distinguish between evaporation and boiling.
- 3.2.7 Define *specific latent heat*.
- 3.2.8 Solve problems involving specific latent heats.

Thermal capacity (C)

If heat is added to a body, its temperature rises, but the actual increase in temperature depends on the body.

The *thermal capacity* (C) of a body is the amount of heat needed to raise its temperature by 1°C . Unit: $\text{J } ^\circ\text{C}^{-1}$

If the temperature of a body increases by an amount ΔT when quantity of heat Q is added, then the thermal capacity is given by the equation:

$$C = \frac{Q}{\Delta T}$$



This applies not only when things are given heat, but also when they lose heat.

Worked example

- 1 If the thermal capacity of a quantity of water is $5000 \text{ J } ^\circ\text{C}^{-1}$, how much heat is required to raise its temperature from 20°C to 100°C ?
- 2 How much heat is lost from a block of metal of thermal capacity $800 \text{ J } ^\circ\text{C}^{-1}$ when it cools down from 60°C to 20°C ?

Solution

- 1 Thermal capacity $Q = \frac{Q}{\Delta T}$ From definition
So $Q = C\Delta T$ Rearranging
Therefore $Q = 5000 \times (100 - 20) \text{ J}$
So the heat required $Q = 400 \text{ kJ}$
- 2 Thermal capacity $C = \frac{Q}{\Delta T}$ From definition
So $Q = C\Delta T$ Rearranging
Therefore $Q = 800 \times (60 - 20) \text{ J}$
So the heat lost $Q = 32 \text{ kJ}$

● **Examiner's hint:** Remember, power is energy per unit time.

Exercises

- 5 The thermal capacity of a 60 kg human is $210 \text{ kJ } ^\circ\text{C}^{-1}$. How much heat is lost from a body if its temperature drops by 2°C ?
- 6 The temperature of a room is 10°C . In 1 hour the room is heated to 20°C by a 1 kW electric heater.
 - (a) How much heat is delivered to the room?
 - (b) What is the thermal capacity of the room?
 - (c) Does all this heat go to heat the room?

Specific heat capacity (c)

The thermal capacity depends on the size of the object and what it is made of. The *specific heat capacity* depends only on the material. Raising the temperature of 1 kg of water requires more heat than raising 1 kg of steel by the same amount, so the specific heat capacity of water is higher than that of steel.

The specific heat capacity of a material is the amount of heat required to raise the temperature of 1 kg of the material by 1°C . Unit: $\text{J kg}^{-1} ^\circ\text{C}^{-1}$

If a quantity of heat Q is required to raise the temperature of a mass m of material by ΔT then the specific heat capacity (c) of that material is given by the following equation:

$$c = \frac{Q}{m\Delta T}$$

The specific heat capacity of water is quite high, so it takes a lot of energy to heat up the water for a shower.



Worked example

- 1 The specific heat capacity of water is $4200 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$. How much heat will be required to heat 300 g of water from 20°C to 60°C ?
- 2 A metal block of mass 1.5 kg loses 20 kJ of heat. As this happens, its temperature drops from 60°C to 45°C . What is the specific heat capacity of the metal?

Solution

1 Specific heat capacity $c = \frac{Q}{m\Delta T}$ From definition

So $Q = cm\Delta T$ Rearranging

Therefore $Q = 4200 \times 0.3 \times 40$ Note: Convert g to kg

$Q = 50.4 \text{ kJ}$

2 Specific heat capacity $c = \frac{Q}{m\Delta T}$ From definition

So $c = 20\,000 / 1.5(60 - 45)$ Rearranging

$c = 888.9 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

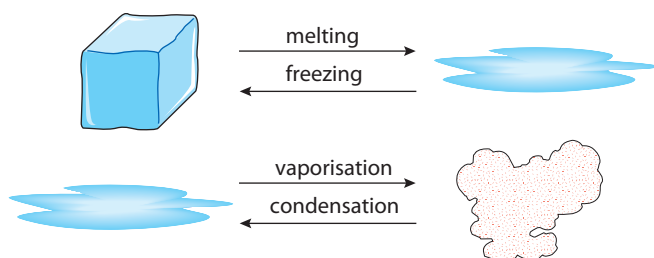
Exercises

Use the data in the table to solve the problems:

Substance	Specific heat capacity ($\text{J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$)
Water	4200
Copper	380
Aluminium	900
Steel	440

- 7 How much heat is required to raise the temperature of 250 g of copper from 20°C to 160°C ?
- 8 The density of water is 1000 kg m^{-3} .
 - (a) What is the mass of 1 litre of water?
 - (b) How much energy will it take to raise the temperature of 1 litre of water from 20°C to 100°C ?
 - (c) A water heater has a power rating of 1 kW. How many seconds will this heater take to boil 1 litre of water?
- 9 A 500 g piece of aluminium is heated with a 500 W heater for 10 minutes.
 - (a) How much energy will be given to the aluminium in this time?
 - (b) If the temperature of the aluminium was 20°C at the beginning, what will its temperature be after 10 minutes?
- 10 A car of mass 1500 kg travelling at 20 m s^{-1} brakes suddenly and comes to a stop.
 - (a) How much KE does the car lose?
 - (b) If 75% of the energy is given to the front brakes, how much energy will they receive?
 - (c) The brakes are made out of steel and have a total mass of 10 kg. By how much will their temperature rise?
- 11 The water comes out of a showerhead at a temperature of 50°C at a rate of 8 litres per minute.
 - (a) If you take a shower lasting 10 minutes, how many kg of water have you used?
 - (b) If the water must be heated from 10°C , how much energy is needed to heat the water?

Change of state

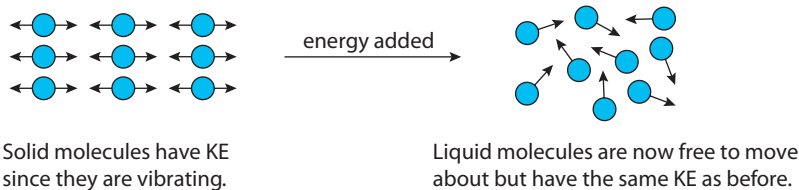


When water boils, this is called a *change of state* (or *change of phase*). As this happens, the temperature of the water doesn't change — it stays at 100°C . In fact,

Figure 3.9 When matter changes from liquid to gas, or solid to liquid, it is changing state.

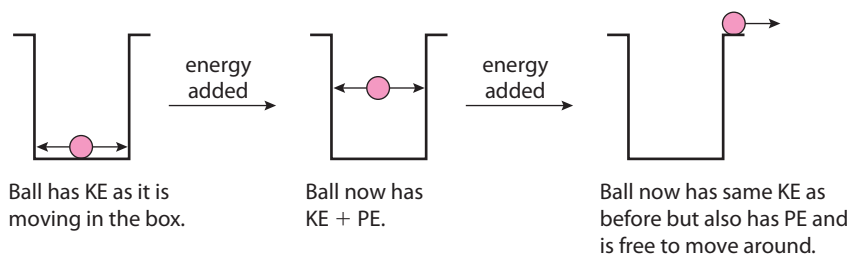
we find that whenever the state of a material changes, the temperature stays the same. We can explain this in terms of the particle model.

Figure 3.10 Molecules gain PE when the state changes.



When matter changes state, the energy is needed to enable the molecules to move more freely. To understand this, consider the example below.

Figure 3.11 A ball-in-a-box model of change of state.



An iceberg melts as it floats into warmer water.



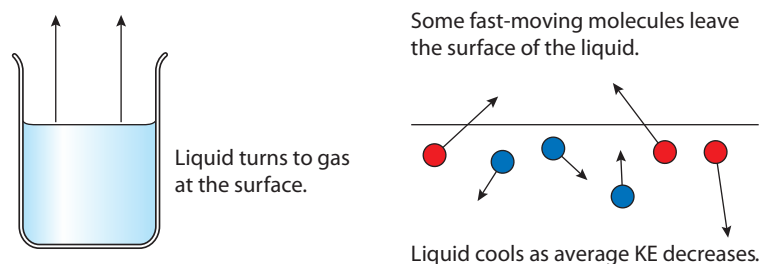
Boiling and evaporation

These are two different processes by which liquids can change to gases.

Boiling takes place throughout the liquid and always at the same temperature.

Evaporation takes place only at the surface of the liquid and can happen at all temperatures.

Figure 3.12 A microscopic model of evaporation.



When a liquid evaporates, the fastest-moving particles leave the surface. This means that the average kinetic energy of the remaining particles is lower, resulting in a drop in temperature.

The rate of evaporation can be increased by:

- Increasing the surface area; this increases the number of molecules near the surface, giving more of them a chance to escape.
- Blowing across the surface. After molecules have left the surface they form a small 'vapour cloud' above the liquid. If this is blown away, it allows further molecules to leave the surface more easily.
- Raising the temperature; this increases the kinetic energy of the liquid molecules, enabling more to escape.

Specific latent heat (L)

The *specific latent heat* of a material is the amount of heat required to change the state of 1 kg of the material without change of temperature.

Unit: J kg^{-1}

Latent means *hidden*. This name is used because when matter changes state, the heat added does not cause the temperature to rise, but seems to disappear.

If it takes an amount of energy Q to change the state of a mass m of a substance, then the specific latent heat of that substance is given by the equation:

$$L = \frac{Q}{m}$$

Worked example

- 1 The specific latent heat of fusion of water is $3.35 \times 10^5 \text{ J kg}^{-1}$. How much energy is required to change 500 g of ice into water?
- 2 The amount of heat released when 100 g of steam turns to water is $2.27 \times 10^5 \text{ J}$. What is the specific latent heat of vaporization of water?

Solution

- 1 The latent heat of fusion $L_f = \frac{Q}{m}$ From definition

So $Q = mL$ Rearranging

Therefore $Q = 0.5 \times 3.35 \times 10^5 \text{ J}$

So the heat required $Q = 1.675 \times 10^5 \text{ J}$

- 2 The specific latent heat of vaporization $L = \frac{Q}{m}$ From definition

Therefore $L = 2.27 \times 10^5 / 0.1 \text{ J kg}^{-1}$

So the specific latent heat of vaporization $L = 2.27 \times 10^6 \text{ J kg}^{-1}$

Exercises

Latent heats of water

Latent heat of vaporization	$2.27 \times 10^6 \text{ J kg}^{-1}$
Latent heat of fusion	$3.35 \times 10^5 \text{ J kg}^{-1}$

Use the data about water in the table to solve the following problems.

- 12 If the mass of water in a cloud is 1 million kg, how much energy will be released if the cloud turns from water to ice?
- 13 A water boiler has a power rating of 800 W. How long will it take to turn 400 g of boiling water into steam?
- 14 The ice covering a 1000 m^2 lake is 2 cm thick.
 - (a) If the density of ice is 920 kg m^{-3} , what is the mass of the ice on the lake?
 - (b) How much energy is required to melt the ice?
 - (c) If the sun melts the ice in 5 hours, what is the power delivered to the lake?
 - (d) How much power does the Sun deliver per m^2 ?

i People sweat to increase the rate at which they lose heat. When you get hot, sweat comes out of your skin onto the surface of your body. When the sweat evaporates, it cools you down. In a sauna there is so much water vapour in the air that the sweat doesn't evaporate.

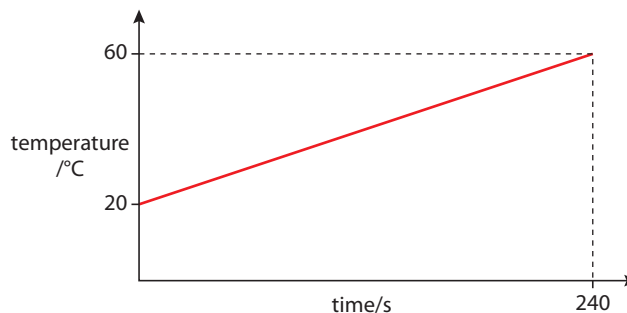
i Solid $\rightarrow$ liquid
Specific latent heat of fusion
Liquid $\rightarrow$ gas
Specific latent heat of vaporization

i This equation ($L = \frac{Q}{m}$) can also be used to calculate the heat lost when a substance changes from gas to liquid, or liquid to solid.

Graphical representation of heating

The increase of the temperature of a body can be represented by a temperature–time graph. Observing this graph can give us a lot of information about the heating process.

Figure 3.13 Temperature–time graph for 1 kg of water being heated in an electric kettle.



In this example, we are ignoring the heat given to the kettle and the heat lost.

From this graph we can calculate the amount of heat given to the water per unit time (power).

$$\text{The gradient of the graph} = \frac{\text{temperature rise}}{\text{time}} = \frac{\Delta T}{t}$$

We know from the definition of specific heat capacity that

$$\text{heat added} = mc\Delta T$$

$$\text{The rate of adding heat} = P = \frac{mc\Delta T}{t}$$

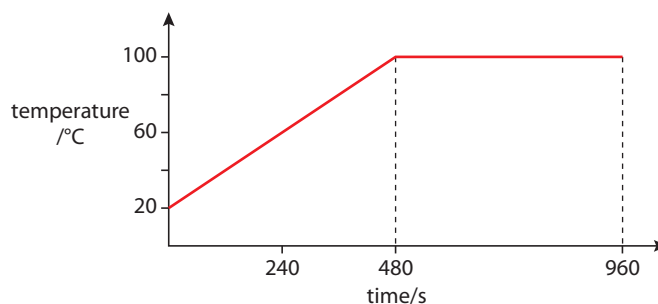
$$\text{So } P = mc \times \text{gradient}$$

$$\text{The gradient of this line} = \frac{(60 - 20)}{240} \text{ °C s}^{-1} = 0.167 \text{ °C s}^{-1}$$

$$\text{So the power delivered} = 4200 \times 0.167 \text{ W} = 700 \text{ W}$$

If we continue to heat this water it will begin to boil.

Figure 3.14 A graph of temperature vs time for boiling water. When the water is boiling, the temperature does not increase any more.



If we assume that the heater is giving heat to the water at the same rate, then we can calculate how much heat was given to the water whilst it was boiling.

$$\text{Power of the heater} = 700 \text{ W}$$

$$\text{Time of boiling} = 480 \text{ s}$$

$$\text{Energy supplied} = \text{power} \times \text{time} = 700 \times 480 \text{ J} = 3.36 \times 10^5 \text{ J}$$

From this we can calculate how much water must have turned to steam.

$$\text{Heat added to change state} = \text{mass} \times \text{latent heat of vaporization,}$$

$$\text{where latent heat of vaporization of water} = 2.27 \times 10^6 \text{ J kg}^{-1}.$$

$$\text{Mass changed to steam} = \frac{3.36 \times 10^5}{2.27 \times 10^6} = 0.15 \text{ kg}$$

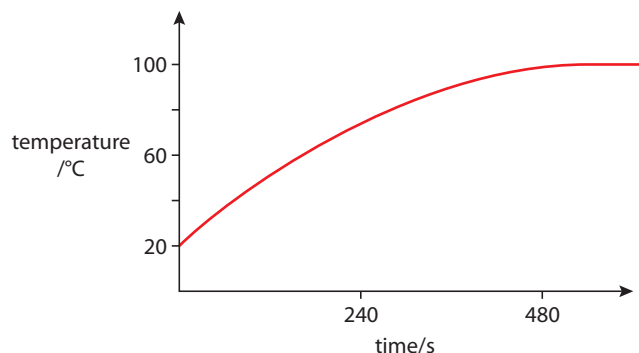


Figure 3.15 Heat loss.



When boiling a kettle, heat is continually being lost to the room. The amount of heat loss is proportional to the temperature of the kettle. For this reason, a graph of temperature against time is actually a curve, as shown in Figure 3.15.

The fact that the gradient decreases, tells us that the amount of heat given to the water gets less with time. This is because as it gets hotter, more and more of the heat is lost to the room.

Measuring thermal quantities by the method of mixtures

The method of mixtures can be used to measure the specific heat capacity and specific latent heat of substances.

Specific heat capacity of a metal

A metal sample is first heated to a known temperature. The most convenient way of doing this is to place it in boiling water for a few minutes; after this time it will be at 100°C . The hot metal is then quickly moved to an insulated cup containing a known mass of cold water. The hot metal will cause the temperature of the cold water to rise; the rise in temperature is measured with a thermometer. Some example temperatures and masses are given in Figure 3.16.

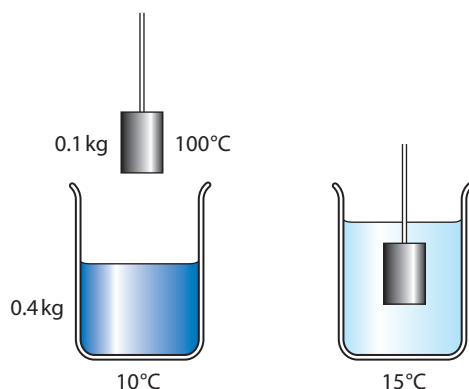


Figure 3.16 Measuring the specific heat capacity of a metal.

As the specific heat capacity of water is $4180 \text{ J kg}^{-1} ^{\circ}\text{C}^{-1}$, we can calculate the specific heat capacity of the metal.

$$\Delta T \text{ for the metal} = 100 - 15 = 85^{\circ}\text{C}$$

$$\text{and } \Delta T \text{ for the water} = 15 - 10 = 5^{\circ}\text{C}$$

Applying the formula $Q = mc\Delta T$ we get

$$(mc\Delta T)_{\text{metal}} = 0.1 \times c \times 85 = 8.5c$$

$$(mc\Delta T)_{\text{water}} = 0.4 \times 4180 \times 5 = 8360$$

If no heat is lost, then the heat transferred from the metal = heat transferred to the water

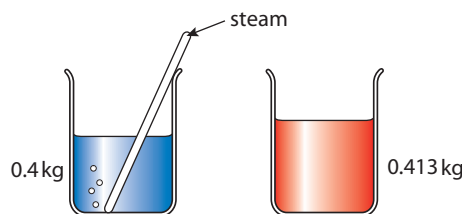
$$8.5c = 8360$$

$$c_{\text{metal}} = 983 \text{ J kg}^{-1} ^{\circ}\text{C}^{-1}$$

Latent heat of vaporization of water

To measure the latent heat of vaporization, steam is passed into cold water. Some of the steam condenses in the water, causing the water temperature to rise. The heat from the steam = the heat to the water.

Figure 3.17 By measuring the rise in temperature, the specific latent heat can be calculated.



In Figure 3.17, 13 g of steam have condensed in the water, raising its temperature by 20 °C. The steam condenses then cools down from 100 °C to 30 °C.

$$\text{Heat from steam} = ml_{\text{steam}} + mc\Delta T_{\text{water}}$$

$$0.013 \times L + 0.013 \times 4.18 \times 10^3 \times 70 = 0.013L + 3803.8$$

$$\begin{aligned} \text{Heat transferred to cold water} &= mc\Delta T_{\text{water}} = 0.4 \times 4.18 \times 10^3 \times 20 \\ &= 33\,440 \text{ J} \end{aligned}$$

Since heat from steam = heat to water

$$0.013L + 3803.8 = 33\,440$$

$$\text{So } L = \frac{33\,440 - 3803.8}{0.013}$$

$$L = 2.28 \times 10^6 \text{ J kg}^{-1}$$

Heat loss

In both of these experiments, some of the heat coming from the hot source can be lost to the surroundings. To reduce heat loss, the temperatures can be adjusted, so you could start the experiment below room temperature and end the same amount above (e.g. if room temperature is 20 °C, then you can start at 10 °C and end at 30 °C).

Transfer of water

In the specific heat capacity experiment, droplets of hot water may be transferred with the metal block. This would add extra energy to the water, causing the temperature to rise a little bit too high. In the latent heat experiment, droplets of water sometimes condense in the tube – since they have already condensed, they don't give so much heat to the water.

3.3

Kinetic model of an ideal gas

Assessment statements

- 3.2.9 Define *pressure*.
- 3.2.10 State the assumptions of the kinetic model of an ideal gas.
- 3.2.11 State that temperature is a measure of the average random kinetic energy of the molecules of an ideal gas.
- 3.2.12 Explain the macroscopic behaviour of an ideal gas in terms of a molecular model.

The ideal gas

Of the three states of matter, the gaseous state has the simplest model; this is because the forces between the molecules of a gas are very small, so they are able to move freely. We can therefore use what we know about the motion of particles learnt in the mechanics section to study gases in more detail.

According to our simple model, a gas is made up of a large number of perfectly elastic, tiny spheres moving in random motion.

This model makes some assumptions:

- The molecules are perfectly elastic.
- The molecules are spheres.
- The molecules are identical.
- There are no forces between the molecules (except when they collide) – this means that the molecules move with constant velocity between collisions.
- The molecules are very small, that is, their total volume is much smaller than the volume of the gas.

Some of these assumptions are not true for all gases, especially when the gas is compressed (when the molecules are so close together that they experience a force between them). The gas then behaves as a liquid. However, to keep things simple, we will only consider gases that behave like our model. We call these gases ideal gases.



Temperature of a gas

From our general particle model of matter, we know that the temperature of a gas is directly related to the average KE of the molecules. If the temperature increases, then the speed of the particles will increase.

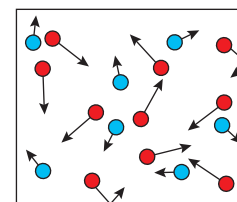
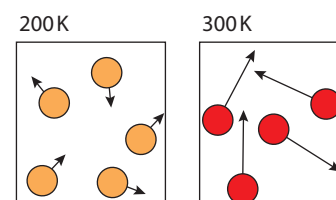


Figure 3.18 Simple model of a gas in a box. In reality the molecules have a range of velocities, not just two.

Nitrogen becomes a liquid at low temperatures.

Figure 3.19 The molecules in a hot gas have a higher average KE.



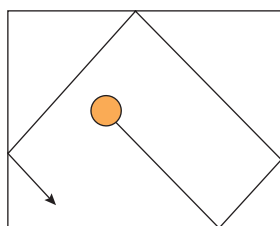


Figure 3.20 A rubber ball bouncing around a box.

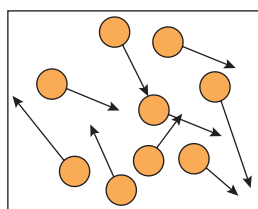


Figure 3.21 Many rubber balls bouncing around a box.

The atmosphere also exerts a pressure; this changes from day to day but is approximately 100 kPa.
1 pascal = 1 Pa = 1 N m⁻²

Figure 3.22 A gas in a piston can be used to vary the properties of a gas.

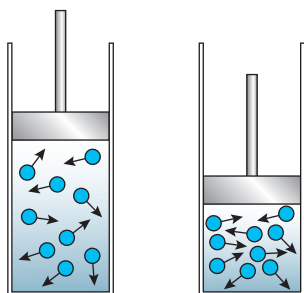


Figure 3.23 The volume of a gas is reduced.

Pressure of a gas

Let us apply what we know about particles to one molecule of a gas. Consider a single gas molecule in a box. According to the model, this is like a perfectly elastic sphere bouncing off the sides

We can see that this particle keeps hitting the walls of the container. Each time it does this, its direction, and therefore its velocity, changes.

Newton's first law of motion says that if a particle isn't at rest or moving with a constant velocity then it must be experiencing an unbalanced force. The particle is therefore experiencing an unbalanced force.

Newton's second law says that the size of this force is equal to the rate of change of momentum, so the force will be greater if the particle travels with a greater speed, or hits the sides more often.

Newton's third law says that if body A exerts a force on body B, then body B will exert an equal and opposite force on A. The wall exerts a force on the particle, so the particle must exert a force on the wall.

If we now add more molecules (as in Figure 3.21) then the particles exert a continuous force, F , on the walls of the container. If the walls have a total area A , then since

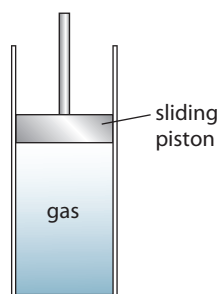
$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

we can say that the pressure exerted on the walls is F/A — in other words, the particles exert a pressure on the container.

It is important to realise that we have been talking about the gas model, not the actual gas. The model predicts that the gas should exert a pressure on the walls of its container and it does.

Properties of a gas

We can now use the particle model to explain why a gas behaves as it does.

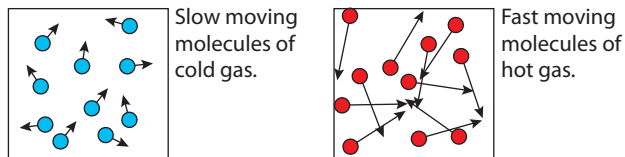


If you push on the piston you can feel the gas push back.

Pressure and volume

If the volume is reduced, the particles hit the walls more often, since the walls are closer together. The force exerted by the molecules is equal to the rate of change of momentum; this will increase if the hits are more frequent, resulting in an increased pressure.

Pressure and temperature



Increase in temperature increases the speed of the molecules. When the molecules hit the walls, their change of momentum will be greater and they will hit the walls more often. The result is a greater rate of change of momentum and hence a larger force. This results in an increase in pressure.

Doing work on a gas

When you push the piston of a pump, it collides with the molecules, giving them energy (rather like a tennis racket hitting a ball). You are doing work on the gas. The increase in kinetic energy results in an increase in temperature and pressure. This is why the temperature of a bicycle pump increases when you pump up the tyres.

Gas does work

When a gas expands, it has to push away the surrounding air. In pushing the air away, the gas does work, and doing this work requires energy. This energy comes from the kinetic energy of the molecules, resulting in a reduction in temperature. This is why an aerosol feels cold when you spray it; the gas expands as it comes out of the canister.

Figure 3.24 The temperature of a gas is increased.



W To understand how pressure, temperature and volume of a gas are related, visit heinemann.co.uk/hotlinks, enter the express code 4426P and click on Weblink 3.2.

3.4 Thermodynamics

Assessment statements

- 10.1.3 Describe the concept of the absolute zero of temperature and the Kelvin scale of temperature
- 10.1.1 State the equation of state for an ideal gas.
- 10.1.4 Solve problems using the equation of state of an ideal gas.
- 10.1.2 Describe the difference between an ideal gas and a real gas.

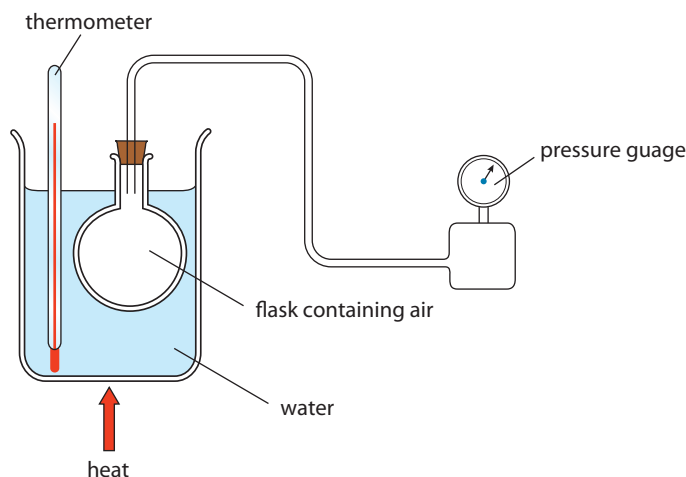


Figure 3.25 The pressure of a fixed volume of gas can be measured as the temperature is changed.

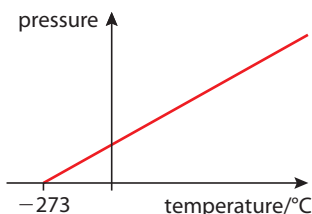


Figure 3.26

The absolute temperature scale (Kelvin)

Absolute zero

If we measure the pressure of a fixed volume of gas at different temperatures we find that temperature and pressure are linearly related as shown in Figure 3.26. We can explain this using the kinetic theory in the following way. Increasing the temperature results in an increase in the average KE of the molecules, so the molecules start to move faster. When the fast moving molecules collide with the walls of the container, the change in momentum is greater so the force exerted on the wall is greater (according to Newton). This, coupled with the fact that the collisions will now occur more often, results in an increase in pressure.

We can see from Figure 3.26 that the line passes through the x -axis at -273°C . This is the temperature when the pressure of the gas is zero. According to the kinetic theory, this is the point at which the molecules have stopped moving, which suggests that there must be a lowest possible temperature. We would not have come to the same conclusion if we had based our temperature measurement on the length of a metal rod since this will never be zero, no matter how cold it is.

Defining the scale

All temperature scales are based on some physical property that changes with changing temperature. The absolute temperature scale is based on the pressure of a fixed mass of gas kept at constant volume. To define the size of any unit we need two fixed points (e.g. the metre could be defined in terms of the two ends of a metal rod). The absolute temperature scale is an exception to this rule, since the zero on the scale is absolute. In this case we only need one fixed point. A fixed point used to define a temperature scale is some observable event that always takes place at the same temperature; the commonly used examples are the melting and boiling of pure water at normal atmospheric pressure. The one used to define the absolute temperature scale is the triple point of water. This is the temperature at which water exists as solid, liquid and gas in equilibrium, 0.01°C , almost the same as the freezing point.

Setting the size of the unit

To set the size of the unit we simply divide the difference between the fixed points into a convenient number of steps. The metre, for example, is divided into 100 centimetre steps. It might seem sensible to split the difference between the triple point and absolute zero into 100; however, this would make converting between Celsius and the new scale rather awkward. It is much better to choose the same division as used in the Celsius scale, then the units will be of equal size. If we look at Figure 3.26 we see that there are 273°C between absolute zero and the triple point (about 0°C), so if we make the triple point 273 on the new scale, the two will be the same.

To clarify this, let us take an example.

Figure 3.27 shows the sort of apparatus that could be used to carry out this experiment. A sample of gas is cooled down to the triple point by placing it in a bath containing water, ice and steam in equilibrium. The pressure of the gas at this temperature is then measured to be 75kPa. To make the scale, this point is

There is no problem having ice and water existing in equilibrium but to make the water boil at the same time, the pressure must be reduced.



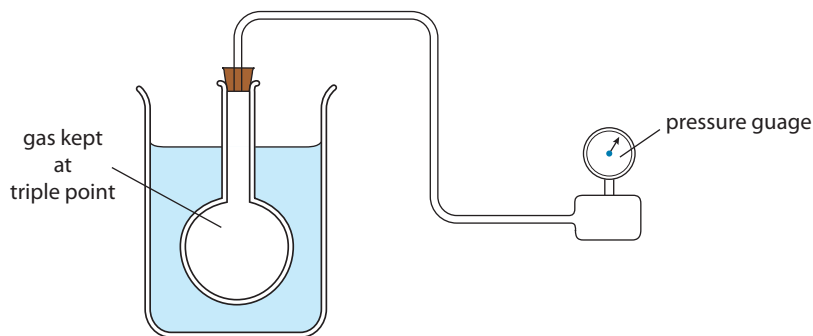


Figure 3.27

plotted on a graph as shown in Figure 3.28. By choosing the triple point to be 273 we make the gradient of this graph the same as Figure 3.26. This makes conversion from Celsius a simple matter of adding 273.

Converting between the Kelvin and Celsius scales

$$0 \text{ K} = -273 \text{ }^{\circ}\text{C}$$

$$273 \text{ K} = 0 \text{ }^{\circ}\text{C}$$

To convert from K to $^{\circ}\text{C}$ subtract 273.

To convert from $^{\circ}\text{C}$ to K add 273.

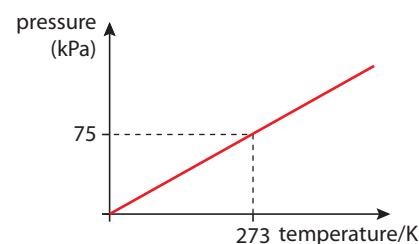
Since the size of the unit is the same, a *change* in temperature is the same in K and $^{\circ}\text{C}$.

Example

Water is heated from $20 \text{ }^{\circ}\text{C}$ to $80 \text{ }^{\circ}\text{C}$; this is a change of $80 - 20 = 60 \text{ }^{\circ}\text{C}$

In kelvin, this is from 293 K to 353 K; this is a change of $353 - 293 = 60 \text{ K}$

Figure 3.28 Graph used to define the Kelvin scale.



For more precise calculations you should use a value of $273.15 \text{ K} = 0^{\circ}\text{C}$

Defining the state of a gas

Defining the state of a body means to give all the quantities necessary to enable someone else to recreate the exact conditions that you observe. A piece of metal can have many different temperatures and when its temperature is increased it expands. So to define the state of a given mass of metal you would need to quote its temperature and volume. To represent these states graphically a graph of temperature against volume could be plotted. All possible states would then lie on a straight line.

To define the state of a sample of gas you need to quote three quantities: pressure, volume and temperature. To represent these three quantities we would need to plot a three-dimensional graph. If we do this for an ideal gas we get a curved surface like the one shown in Figure 3.29. All possible states of the gas then lie on the curved surface.

The equation of this surface is

$$PV = nRT$$

where n = the number of moles

R = the molar gas constant ($8.31 \text{ J mol}^{-1} \text{ K}^{-1}$)

This is called the equation of state for an ideal gas.

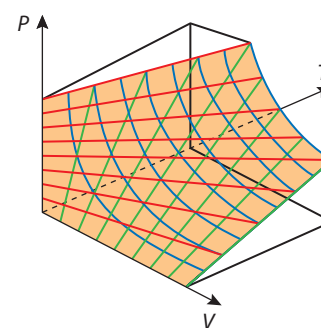


Figure 3.29

Worked example

The pressure of a gas inside a cylinder is 300 kPa. If the gas is compressed to half its original volume and the temperature rises from $23 \text{ }^{\circ}\text{C}$ to $323 \text{ }^{\circ}\text{C}$, what will its new pressure be?

Solution

Using the ideal gas equation: $PV = nRT$

Rearranging: $\frac{PV}{T} = \text{constant}$

So $\frac{PV}{T}$ at the beginning = $\frac{PV}{T}$ at the end

$$\frac{PV}{T} \text{ at the beginning} = \frac{300\,000 \times V}{300}$$

$$\frac{PV}{T} \text{ at the end} = \frac{P \times V/2}{600}$$

$$\begin{aligned} \text{Equating: } 300\,000 \times \frac{V}{300} &= \frac{P \times V/2}{600} \\ P &= 300\,000 \times 600 \times \frac{2}{300} \\ P &= 1200 \text{ kPa} \end{aligned}$$

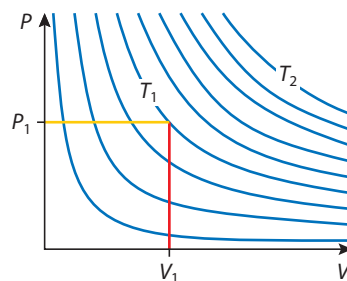
Exercises

- 15 The pressure of 10 m^3 of gas in a sealed container at 300 K is 250 kPa . If the temperature of the gas is changed to 350 K , what will the pressure be?
- 16 A container of volume 2 m^3 contains 5 moles of gas. If the temperature of the gas is 293 K .
 - (a) what is the pressure exerted by the gas?
 - (b) what is the new pressure if half of the gas leaks out?
- 17 A piston contains 250 cm^3 of gas at 300 K and a pressure of 150 kPa . The gas expands, causing the pressure to go down to 100 kPa and the temperature drops to 250 K . What is the new volume?
- 18 A sample of gas trapped in a piston is heated and compressed at the same time. This results in a doubling of temperature and a halving of the volume. If the initial pressure was 100 kPa , what will the final pressure be?

PV diagrams

It is rather difficult to draw the 3D graph that represents PV and T so to make life simpler we draw the view looking along the T axis. This is called a PV diagram and is shown in Figure 3.30.

Figure 3.30 The curved surface of Figure 3.29 viewed in 2D.



You can imagine that the whole area of the PV diagram is covered in points. Each point represents the gas at a different state, in other words with different PV and T . If we join up all the points with equal pressure P_1 we get a horizontal line; if we join all points with equal volume V_1 we get a vertical line, and joining all points with equal temperature T_1 gives the blue curve shown.

The higher temperature curves are the ones further away from the origin, so $T_2 > T_1$. To explain this, consider a gas at volume V_1 pressure P_1 and temperature T_1 . If the volume is kept constant as the temperature is increased, its pressure will increase. So its position on the diagram will move up the y -axis to a higher blue

line. The blue lines are called isotherms, and although they are not always drawn on the PV diagram you must never forget that they are there.

Gas transformations

A gas can be heated, cooled, compressed and expanded. But, according to the equation of state for an ideal gas, whatever we do to the gas the value of $\frac{PV}{T}$ will remain the same. We can represent these changes on a PV diagram as illustrated in Figure 3.31.

Constant pressure (isobaric)

The line A–B represents a constant pressure change. From A to B the volume is getting smaller, so this is a compression. When this happens, we can also deduce that the temperature must decrease, since the gas moves to a lower isotherm.

Constant volume (isochoric)

The line B–C represents a constant volume change. From B to C the pressure is increasing. This is because the temperature is increasing as can be deduced from the fact that the gas is changing to a higher temperature isotherm.

Constant temperature (isothermal)

The line C–A is an isotherm so represents a change at constant temperature. From C to A the volume is increasing, so this is an expansion.

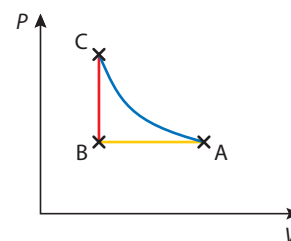


Figure 3.31

Exercise

- 19 Figure 3.32 represents four different transitions performed on the same sample of a gas. For each transition (a, b, c and d) deduce whether each of P , V and T go up, down or stay the same.

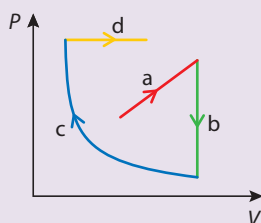


Figure 3.32

Real gases

If we compress a gas, the molecules get closer and closer together.

According to our simple kinetic theory there is no force between the molecules, so the only effect of compressing the gas is that the molecules become denser and therefore hit the walls more often, resulting in a proportional increase in pressure. However, as the gas molecules become very close to each other, the force between them is no longer negligible. As you try to push them closer, they push back. The pressure is no longer proportional to the volume but rises very steeply as the volume is reduced. What has happened is that the gas has changed to a liquid and that is what most real gases do, unless the temperature is very high. At high temperatures, no matter how much you compress the gas it will not turn to liquid.

i Ideal gases and real gases

Ideal and real are not two types of gas. Ideal and real are the way the gas behaves. P , V and T for an ideal gas are related by the equation $PV = nRT$. Carbon dioxide behaves like an ideal gas at high temperatures and low pressures but not at low temperatures and high pressures.

3.5 Thermodynamic processes

Assessment statements

- 10.2.1 Deduce an expression for the work involved in a volume change of a gas at constant pressure.
- 10.2.2 State the first law of thermodynamics.
- 10.2.3 Identify the first law of thermodynamics as a statement of the principle of energy conservation.
- 10.2.4 Describe the isochoric (isovolumetric), isobaric, isothermal and adiabatic changes of state of an ideal gas.
- 10.2.5 Draw and annotate thermodynamic processes and cycles on P - V diagrams.
- 10.2.6 Calculate from a P - V diagram the work done in a thermodynamic cycle.
- 10.2.7 Solve problems involving state changes of a gas.

Thermodynamic system

Thermodynamics relates to a thermodynamic system – this is a collection of bodies that can do work on and exchange heat between each other. The car engine and the human body are both thermodynamic systems. In this course we will consider only the simple system of a gas trapped by a piston. However, the laws apply to all systems.

Energy and gas transformations

When dealing with the motion of a simple ball, we found that if we used the law of conservation of energy to solve problems, it was often simpler than going into the details of all the forces, acceleration etc. The same is true when dealing with the billions of particles that make up a gas – as long as the system is isolated, we can use the conservation of energy to predict the outcome of any transformation. But before we can do that, we need to know in what way energy is involved when a gas changes state.

Internal energy

From the study of mechanics we know that a particle can possess two types of energy: PE and KE. According to our simple kinetic model of a gas there are no forces between the molecules. This means that to move a molecule around requires no work to be done (work done = force $\times$ distance) so moving the molecules will not result in a change in the PE. On the other hand, we know that the molecules are moving about in random motion – they therefore do possess KE. The sum of all the KE of all the molecules is called the *internal energy*.

Work done

Work is done when the point of application of a force moves in the direction of the force. If the pressure of a gas pushes a piston out, then the force exerted on the piston is moving in the direction of the force, so work is done. The example in Figure 3.33 is of a gas expanding at constant pressure. In this case, the force exerted on the piston = $P \times A$. The work done when the piston moved distance Δd is therefore given by:

$$\text{Work done} = P \times A \times \Delta d$$

but $A\Delta d$ is the change in volume ΔV , so

$$\text{Work done} = P\Delta V$$

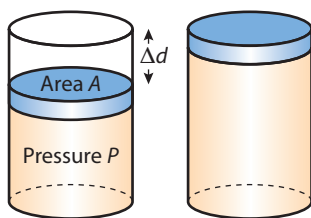
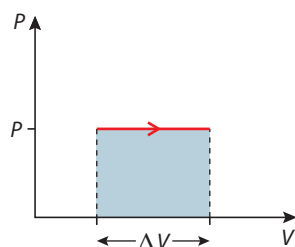


Figure 3.33 A gas expands at constant pressure.

Figure 3.34 is the PV graph for this constant pressure expansion. From this we can see that the work done is given by the area under the graph. This is true for all processes.



Heat

Heat is the name given to the energy that is exchanged when a hot body is in contact with a cold one. Adding heat to a gas can increase its internal energy and allow it to do work.

The first law of thermodynamics

According to the law of conservation of energy, energy can neither be created nor destroyed, so the amount of heat, Q , added to a gas must equal the work done by the gas, W , plus the increase in internal energy, ΔU . This is so fundamental to the way physical systems behave that it is called the *first law of thermodynamics*. This can be written in the following way

$$Q = \Delta U + W$$

This would be nice and easy if the only thing a gas could do is gain heat, get hot and do work. However, heat can be added and lost, work can be done by the gas and on the gas and the internal energy can increase and decrease. To help us understand all the different possibilities, we will use the PV diagram to represent the states of a gas.

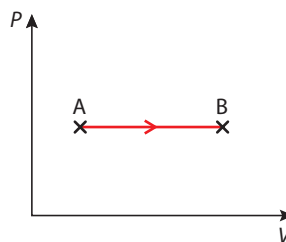
Using PV diagrams in thermodynamics

We have seen how a PV diagram enables us to see the changes in P , V and T that take place when a gas changes from one state to another. It also tells us what energy changes are taking place. If we consider the transformation represented in Figure 3.35 we can deduce that when the gas changes from A to B:

- 1 Since the volume is increasing, the gas is doing work (W is positive).
- 2 Since the temperature is increasing, the internal energy is increasing (ΔU is positive).

If we then apply the first law $Q = \Delta U + W$ we can conclude that if both ΔU and W are positive then Q must also be positive, so heat must have been added.

This is a typical example of how we use the PV diagram with the first law; we use the diagram to find out how the temperature changes and whether work is done *by* the gas or *on* the gas, and then use the first law to deduce whether heat is added or lost.



Sign of work

When a gas does work, it is pushing the piston out; this is positive.

If work is done on the gas then something must be pushing the piston in. This is negative work.

Figure 3.34



First law (simple version)

If a gas expands and gets hot, heat must have been added.

Figure 3.35

Examiner's hint:

Change in **volume** tells us whether work is done by the gas or on it.

Change in **temperature** tells us whether the internal energy goes up or down.

Change in **pressure** is not interesting.

Examples

Constant pressure contraction (isobaric)

The example of Figure 3.35 was an isobaric expansion; if we reverse this transformation ($B \rightarrow A$) we get an isobaric compression. In this case:

- 1 Temperature decrease implies that the internal energy decreases ($\Delta U = \text{negative}$).
- 2 Volume decrease implies that work is done on the gas ($W = \text{negative}$).

Applying the first law, $Q = \Delta U + W$, tells us that Q is also negative, so heat is lost.

Constant volume increase in temperature (isochoric)

Figure 3.36 is the PV graph for a gas undergoing a constant volume transformation. From the graph we can deduce that:

- 1 The volume isn't changing, so no work is done ($W = 0$).
- 2 The gas changes to a higher isotherm so the temperature is increasing; this means that the internal energy is increasing ($\Delta U = \text{positive}$).

Applying the first law $Q = \Delta U + W$ we can conclude that $Q = \Delta U$ so if ΔU is positive then Q is also positive – heat has been added.

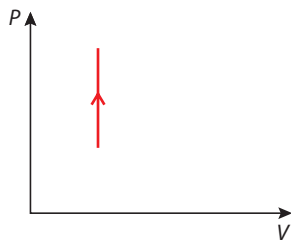


Figure 3.36 An isochoric transformation.

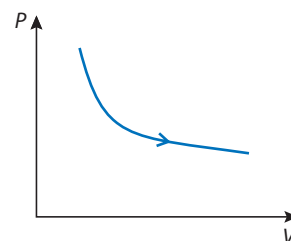


Figure 3.37 A isothermal expansion.

Isothermal expansion

When a gas expands isothermally the transformation follows an isotherm as shown in Figure 3.37. From this PV diagram we can deduce that:

- 1 The temperature doesn't change so there is no change in internal energy ($\Delta U = 0$).
- 2 The volume increases so work is done by the gas ($W = \text{positive}$).

Applying the first law, $Q = \Delta U + W$, we conclude that $Q = W$ so heat must have been added. The heat added enables the gas to do work.

Adiabatic contraction

An adiabatic transformation is one where no heat is exchanged ($Q = 0$). This transformation is represented by the red line in Figure 3.38. It is much steeper than an isothermal. From the PV diagram we can deduce that:

- 1 The volume is reduced so work is done on the gas ($W = \text{negative}$).
- 2 The temperature increases so the internal energy increases ($\Delta U = \text{positive}$).

We also know that $Q = 0$ so if we apply the first law $Q = \Delta U + W$, we get

$$0 = \Delta U - W$$

Rearranging gives $\Delta U = W$, so the work done on the gas goes to increase the internal energy.

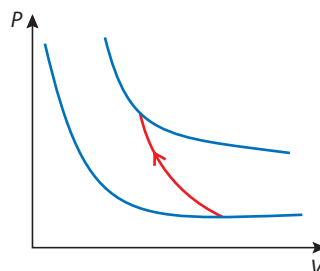


Figure 3.38 Adiabatic contraction.

Exercises

- 20** Calculate the work done by the gas when it expands from a volume of 250 cm^3 to 350 cm^3 at a constant pressure of 200 kPa .
- 21** Estimate the work done by the gas that undergoes the transformation shown in Figure 3.39.

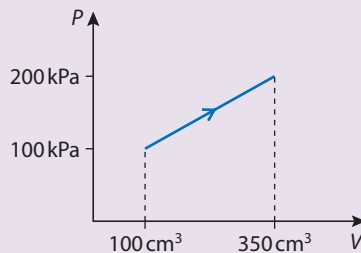


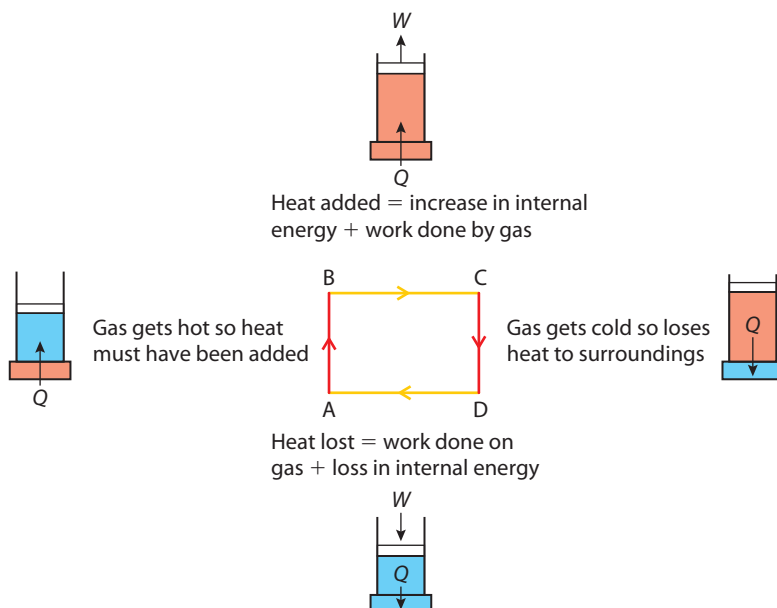
Figure 3.39

Thermodynamic cycles

A thermodynamic cycle is represented by a closed loop on a PV diagram as in Figure 3.40. In this example the cycle is clockwise so the sequence of transformations is:

- A–B isochoric temperature rise
- B–C isobaric expansion
- C–D isochoric temperature drop
- D–A isobaric compression.

In the process of completing this cycle, work is done on the gas from D to A and the gas does work from B to C. It is clear from the diagram that the work done by the gas is greater than the work done on the gas (since the area under the graph is greater from B to C than from D to A) so net work is done. What we have here is an engine; heat is added and work is done. Let us look at this cycle more closely.



Examiner's hint:

You can't easily tell if a curve on a PV graph is adiabatic or isothermal - you have to be told by the examiner. If you have both on a diagram then the adiabatic is the steeper one.

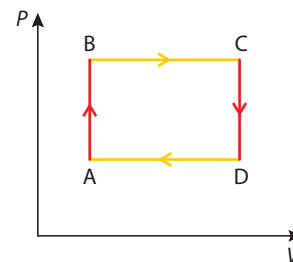


Figure 3.40 A thermodynamic cycle.

Figure 3.41 An example of a thermodynamic cycle, the red and blue rectangles placed under the piston represent hot and cold bodies used to add and take away heat.



Net work done

The net work done during a cycle is the difference between the work done by the gas and the work done on the gas. This is equal to the area enclosed by the cycle on the PV diagram.

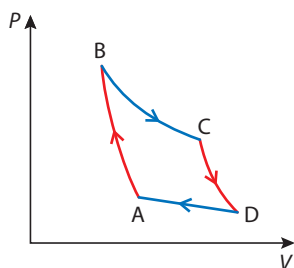
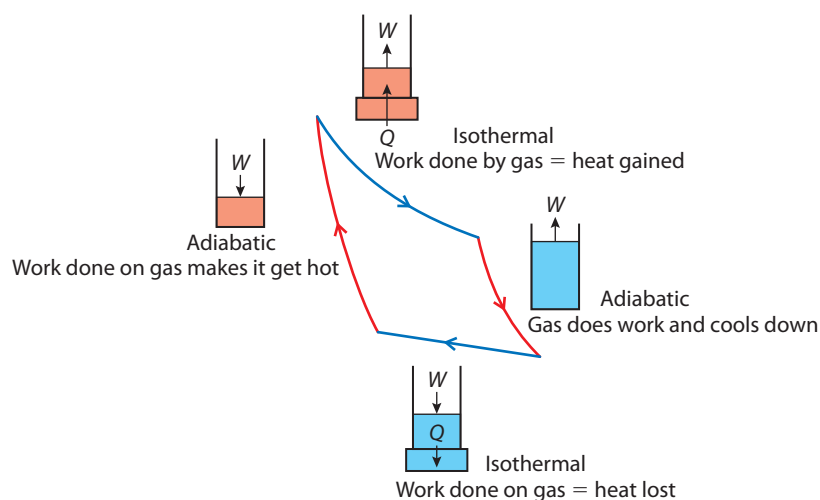


Figure 3.42 The Carnot cycle.

Figure 3.43 The Carnot cycle in detail.

It is much easier to visualise what is happening if you look at a simulation. To do this, visit www.heinemann.co.uk/hotlinks, enter the express code 4426P and click on Weblink 3.3.



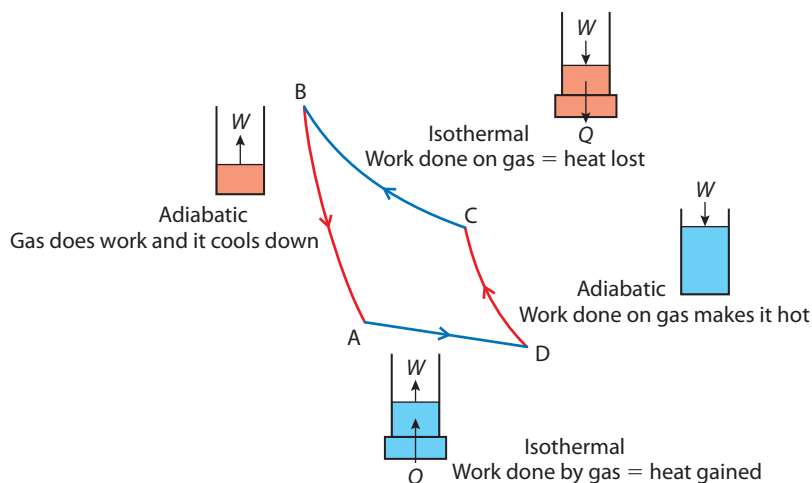
The reverse cycle

Let us consider what would happen if the Carnot cycle was operated in reverse. The details of this are shown in Figure 3.44

Figure 3.44 The reverse Carnot cycle.

The heat pump

A heat pump is used to extract heat from the cold air outside and give it to the inside of a house. It works in exactly the same way as a refrigerator.



The interesting thing about this cycle is that heat is lost to the hot body during the isothermal compression (C to B) and gained from the cold body during the isothermal expansion (A to D). So heat has been taken from something cold and given to something hot. This is what a refrigerator does - it takes heat from the cold food inside and gives it to the warm room. To make this possible, work must be done on the gas (D to C) so that it gets hot enough to give heat to the hot body.

Exercises

22 250 cm³ of gas at 300 K exerts a pressure of 100 kPa on its container; call this state A. It undergoes the following cycle of transformations:

- (i) an isobaric expansion to 500 cm³ (state B)
 - (ii) an isochoric transformation to a pressure of 200 kPa (state C)
 - (iii) an isobaric contraction back to 250 cm³ (state D)
 - (iv) an isochoric transformation back to state A.
- (a) Sketch a PV diagram representing this cycle, labelling the states A, B, C and D.
 (b) Use the ideal gas equation to calculate the temperature at B, C and D.
 (c) Calculate the amount of work done by the gas.
 (d) Calculate the amount of work done on the gas.
 (e) What is the net work done during one cycle?

23 Figure 3.45 represents a Carnot cycle. The areas of the coloured regions are as follows

- A – 50 J
- B – 45 J
- C – 40 J
- D – 35 J
- E – 150 J

If the cycle is performed clockwise, how much work is done:

- (a) during the isothermal expansion?
- (b) during the adiabatic compression?
- (c) by the gas?
- (d) on the gas?
- (e) in total?

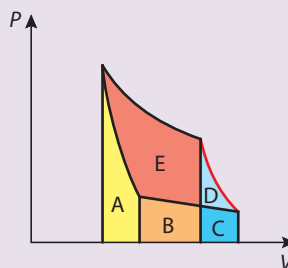


Figure 3.45

3.6

The second law of thermodynamics

Assessment statements

- 10.3.1 State that the second law of thermodynamics implies that thermal energy cannot spontaneously transfer from a region of low temperature to a region of high temperature.
- 10.3.2 State that entropy is a system property that expresses the degree of disorder in the system.
- 10.3.3 State the second law of thermodynamics in terms of entropy changes.
- 10.3.4 Discuss examples of natural processes in terms of entropy changes.

Isothermal expansions

When looking at examples of the first law of thermodynamics, we used isothermal transformations as a simple example. However, if we look at these processes closely we find that they are not possible. Consider a gas enclosed by a piston shown in Figure 3.46. To enable this to expand we could put the gas in contact with a hot source. Heat would then flow into the gas increasing the KE of the molecules. The increased pressure would then push the cylinder out doing work against the surroundings, and the work done would cause the temperature of the gas to drop. For this expansion to be isothermal the temperature would have to stay constant.

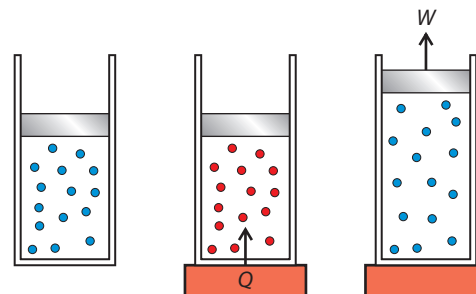
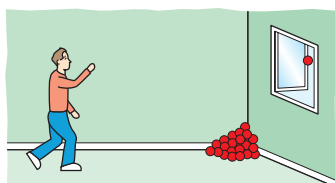
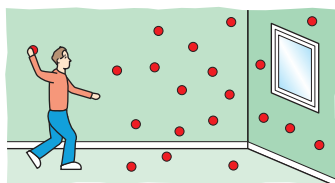


Figure 3.46



The energy you give to the gas must be directly transferred to the piston without increasing the average KE of the gas. The problem is that when the energy is given to the gas molecules, the energy spreads out; this is because all the gas molecules collide into one another. Once the energy is spread out it is impossible to get it all back again. Imagine you are in a room full of perfectly elastic red balls that are bouncing around the room in random motion. You are standing on one side of the room and on the other side is a window. You want to open the window by throwing one of the balls at it. The problem is that every time you throw the ball it hits all the others, and the KE you give it is shared amongst all the other balls and it never reaches its destination.

According to the first law, as long as energy is conserved, anything is possible; but now we see that some things aren't possible. So, to take into account the fact that some things aren't possible, we add the second law:

It is not possible to convert heat completely into work.

Absolute zero

There is one solution to this problem and you can work it out by considering the situation in the room with the rubber balls. If you catch all the rubber balls and put them on the floor you could then take one of the balls and throw it at the window. It would now be able to fly across the room uninterrupted and push the window open. In terms of the gas, this is equivalent to reducing the temperature to zero kelvin, so according to our model, an isothermal expansion is only possible at 0 K. The problem of doing this in the room is that every time you catch a ball and put it down it would get hit by one of the others. You would never be able to stop all the balls, and for the same reason it is not possible to reach 0 K.

Implications of the second law

The second law does not only tell us that isothermal processes cannot take place but it tells us something fundamental about how the physical world behaves. Here are some examples.

Heat flows from hot to cold

If you have some hot gas next to a cold gas, the heat will always travel from hot to cold, never the other way round. We can explain this by using the kinetic model; the gas is made up of randomly moving particles that collide with each other, and the hot gas has faster moving particles. When a fast particle hits a slow one, energy is transferred from the fast one to the slow one, not the other way around.

All the air in a room never goes out of the window

If you are sitting in a room with the window open, is it possible that suddenly all the air molecules could fly out of the window leaving you in a room with no air? Again, if we consider the kinetic model, all the particles are moving in random motion, so for all the molecules to move out of the window, something would have to push them in the same direction. Without that external force this cannot happen.

Spreading out of energy

An alternative way of quoting the second law is to say that energy always spreads out. This can be illustrated by considering molecules injected into a container with identical velocity, as shown in Figure 3.47.

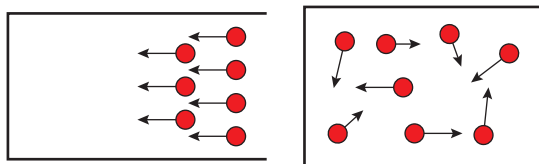


Figure 3.47 Molecules start ordered but end up in random motion.

After they have hit the walls they start to collide with each other. Once this happens, the energy of the individual molecules changes; some will gain energy, some will lose energy, and their motion changes from ordered to disordered. They will never have the same energy again. This is the way of the universe; energy always spreads out.

Entropy

The second law of thermodynamics is about the spreading out of energy. This can be quantified by using the quantity *entropy*.

The change of entropy is ΔS , when a quantity of heat flow into a body at temperature T is equal to $\frac{Q}{T}$.

$$\Delta S = \frac{Q}{T}$$

The unit of entropy is J K^{-1} .

For example, consider the situation of a 1 kg block of ice melting in a room that is at a constant temperature 300 K. To melt the block of ice, it must gain $3.35 \times 10^5 \text{ J}$ of energy. Ice melts at a constant 273 K so:

The gain in entropy of the ice = $\frac{3.35 \times 10^5}{273} = 1.23 \times 10^3 \text{ J K}^{-1}$

The loss of entropy by the room = $\frac{3.35 \times 10^5}{300} = 1.12 \times 10^3 \text{ J K}^{-1}$

We can see from this that the entropy has increased.

Entropy always increases in any transfer of heat since heat always flows from hot bodies to cold bodies. We can therefore rewrite the second law in terms of entropy.

In any thermodynamic process the total entropy always increases.

Entropy is a measure of how spread out or disordered the energy has become. Saying entropy has increased implies that the energy has become more spread out.

Examples

Even though locally entropy can decrease, the total entropy of a system must always increase. Sometimes it's not so easy to see this, but remember it must always happen.



To view the simulation 'gas properties', visit www.heinemann.co.uk/hotlinks, enter the express code 4426P and click on Weblink 3.4.

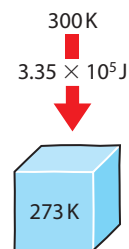


Figure 3.48



The second law is a way of describing the way the bodies in the universe behave. It is often said that something can't happen because the second law says so, but this isn't really true. Some things don't happen because the physical world simply isn't like that, not because the law doesn't allow it.

1 A falling stone

When a stone is held above the ground, all the molecules in the stone are lifted to approximately the same height and therefore have the same PE; this energy is very ordered. If the stone is dropped and hits the ground, this PE is converted into thermal energy in the stone and ground. Each molecule will gain some PE and some KE in a fairly random fashion. The energy is now disordered, and entropy has increased. As time goes on, entropy continues to increase, as this thermal energy spreads out in the ground. It is not possible to collect all this energy back into the stone and have it thrown back into the air; this would be against the second law.

2 The refrigerator

The food in a refrigerator gets cold. As things get cold, the molecules become more ordered, and the entropy in the fridge therefore decreases. According to the second law, entropy must always increase, so the room must gain heat.

3 The petrol engine

In a petrol engine, petrol is burnt to produce heat, which causes a gas to expand, enabling it to do work. If the engine is used to lift a load then the disordered energy in the hot gas has been converted to ordered energy in the lifted load. According to the second law, entropy must increase, so there must be some disorder created. When any engine is used, some heat is lost to the surroundings, and this is where the disorder is created. From this we can deduce that it is impossible to make an engine that does not give heat to something cold; this means an engine can never be 100% efficient.

Exercise

24 500 J of heat flows from a hot body at 400 K to a colder one at 250 K.

- (a) Calculate the entropy change in
 - (i) the hot body
 - (ii) the cold body.
- (b) What is the total change in entropy?

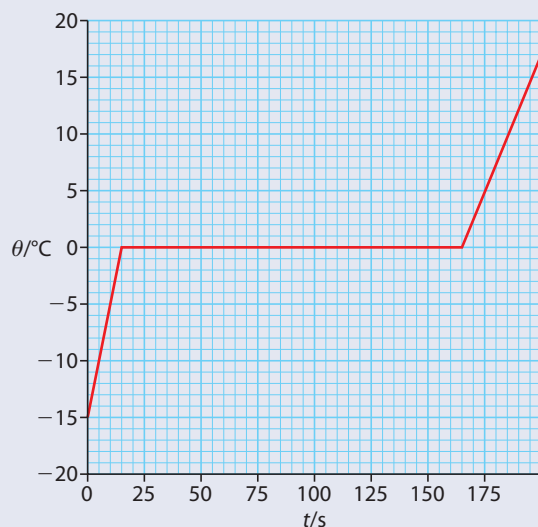
25 Use the second law to explain why heat is released when an electric motor is used to lift a heavy load.

Practice questions**1** This question is about the change of phase (state) of ice.

A quantity of crushed ice is removed from a freezer and placed in a calorimeter. Thermal energy is supplied to the ice at a constant rate. To ensure that all the ice is at the same temperature, it is continually stirred. The temperature of the contents of the calorimeter is recorded every 15 seconds.

The graph at the top of the next page shows the variation with time t of the temperature θ of the contents of the calorimeter. (Uncertainties in the measured quantities are not shown.)

- (a) On the graph, mark with an X, the data point on the graph at which all the ice has just melted. (1)
- (b) Explain, with reference to the energy of the molecules, the constant temperature region of the graph. (3)



The mass of the ice is 0.25 kg and the specific heat capacity of water is $4200 \text{ J kg}^{-1} \text{ K}^{-1}$.

- (c) Use these data and data from the graph to
- (i) deduce that energy is supplied to the ice at the rate of about 530 W (3)
 - (ii) determine the specific heat capacity of ice (3)
 - (iii) determine the specific latent heat of fusion of ice. (2)

2 This question is about thermal physics.

- (a) Explain why, when a liquid evaporates, the liquid cools unless thermal energy is supplied to it. (3)
- (b) State two factors that cause an increase in the rate of evaporation of a liquid. (2)
- (c) A mass of 350 g of water at a temperature of 25°C is placed in a refrigerator that extracts thermal energy from the water at a rate of 86 W. Calculate the time taken for the water to become ice at -5.0°C . (3)

Specific heat capacity of ice = $2.1 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Specific heat capacity of water = $4.2 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Specific latent heat of fusion of ice = $3.3 \times 10^5 \text{ J kg}^{-1}$

3 Explain, in terms of the behaviour of the molecules of an ideal gas, why the pressure of the gas rises when it is heated at constant volume. (3)

4 This question is about the phase (state) changes of the element lead.

A sample of lead has a mass of 0.50 kg and a temperature of 27°C . Energy is supplied to the lead at the rate of 1.5 kW. After 0.2 minutes of heating it reaches its melting point temperature of 327°C . After heating for a further 3 minutes, all the lead has become liquid.

- (a) Assuming that all the energy goes into heating the lead, calculate a value for the
- (i) specific heat capacity of lead (3)
 - (ii) latent heat of fusion of lead. (2)
- (b) Energy continues to be supplied to the lead. Sketch a graph to show how the temperature of the lead varies with time from the start of heating to some 5 minutes after the time when all the lead has become liquid. Indicate on the graph the time at which it starts to melt and the time when it has become liquid. (2)
- (You are not expected to have accurate scales; this is just a sketch graph.)



5 This question is about modelling the thermal processes involved when a person is running. When running, a person generates thermal energy but maintains approximately constant temperature.

(a) Explain what *thermal energy* and *temperature* mean. Distinguish between the two concepts. (4)

The following simple model may be used to estimate the rise in temperature of a runner, assuming no thermal energy is lost.

A closed container holds 70 kg of water, representing the mass of the runner. The water is heated at a rate of 1200 W for 30 minutes. This represents the energy generation in the runner.

(b) (i) Show that the thermal energy generated by the heater is 2.2×10^6 J. (2)

(ii) Calculate the temperature rise of the water, assuming no energy losses from the water. (3)

The specific heat capacity of water is $4200 \text{ J kg}^{-1} \text{ K}^{-1}$.

(c) The temperature rise calculated in (b) would be dangerous for the runner. Outline three mechanisms, other than evaporation, by which the container in the model would transfer energy to its surroundings. (6)

(d) A further process by which energy is lost from the runner is the evaporation of sweat.

(i) Describe, in terms of molecular behaviour, why evaporation causes cooling. (3)

(ii) Percentage of generated energy lost by sweating: 50%

Specific latent heat of vaporization of sweat: $2.26 \times 10^6 \text{ J kg}^{-1}$

Using the information above, and your answer to (b) (i), estimate the mass of sweat evaporated from the runner. (3)

(iii) State and explain two factors that affect the rate of evaporation of sweat from the skin of the runner. (4)

6 A gas is contained in a cylinder fitted with a piston as shown below.



When the gas is compressed rapidly by the piston, its temperature rises because the molecules of the gas

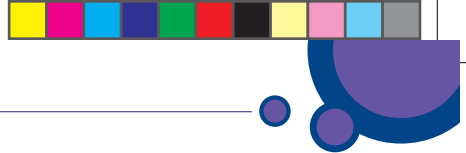
- A are squeezed closer together.
- B collide with each other more frequently.
- C collide with the walls of the container more frequently.
- D gain energy from the moving piston. (1)

7 The Kelvin temperature of an ideal gas is a measure of the

- A average speed of the molecules.
- B average momentum of the molecules.
- C average kinetic energy of the molecules.
- D average potential energy of the molecules. (1)

8 The temperature of an ideal gas is reduced. Which one of the following statements is true?

- A The molecules collide with the walls of the container less frequently.
- B The molecules collide with each other more frequently.
- C The time of contact between the molecules and the wall is reduced.
- D The time of contact between molecules is increased. (1)



9 When a gas in a cylinder is compressed at constant temperature by a piston, the pressure of the gas increases. Consider the following three statements.

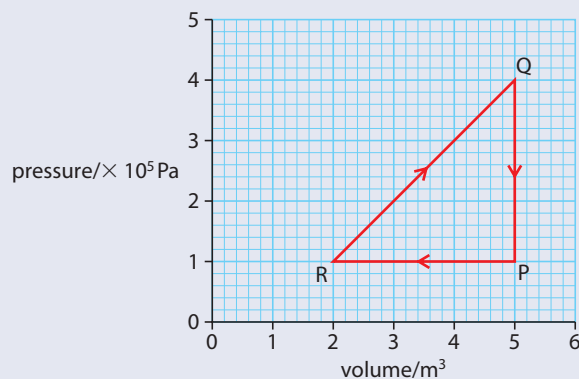
- I The rate at which the molecules collide with the piston increases.
- II The average speed of the molecules increases.
- III The molecules collide with each other more often.

Which statement(s) correctly explain the increase in pressure?

- A I only
- B II only
- C I and II only
- D I and III only

(1)

10 The graph below shows the variation with volume of the pressure of a system.



The work done in compressing the gas from R to P is

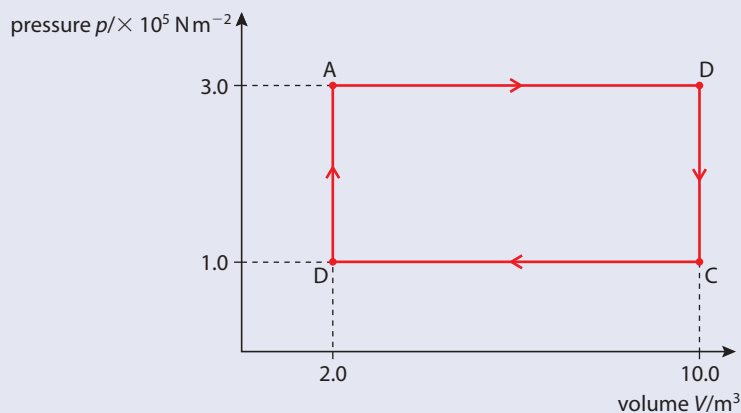
- A $5.0 \times 10^5 \text{ J}$.
- B $4.5 \times 10^5 \text{ J}$.
- C $3.0 \times 10^5 \text{ J}$.
- D 0.

(1)

© International Baccalaureate Organisation

11 This question is about the thermodynamics of a heat engine.

In an idealized heat engine, a fixed mass of a gas undergoes various changes of temperature, pressure and volume. The p - V cycle ($A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$) for these changes is shown in the diagram below.



- (a) Use the information from the graph to calculate the work done during **one** cycle. (2)
- (b) During one cycle, a total of $1.8 \times 10^6 \text{ J}$ of thermal energy is ejected into a cold reservoir. Calculate the efficiency of this engine. (2)



- (c) Using the axes below, sketch the p – V changes that take place in the fixed mass of an ideal gas during one cycle of a Carnot engine. (Note this is a sketch graph – you do not need to add any values.)



- (d) (i) State the names of the **two** types of change that take place during one cycle of a Carnot engine. (2)
- (ii) Add labels to the above graph to indicate which parts of the cycle refer to which particular type of change. (2)

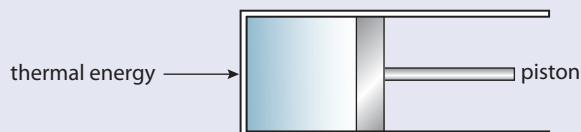
(Total 10 marks)

© International Baccalaureate Organisation

12 This question is about thermodynamic processes.

- (a) Distinguish between an *isothermal* process and an *adiabatic* process as applied to an ideal gas. (2)

An ideal gas is held in a container by a moveable piston and thermal energy is supplied to the gas such that it expands at a constant pressure of 1.2×10^5 Pa.



The initial volume of the container is 0.050 m^3 and after expansion the volume is 0.10 m^3 . The total energy supplied to the gas during the process is $8.0 \times 10^3 \text{ J}$.

- (b) (i) State whether this process is **either** isothermal **or** adiabatic **or** neither. (1)
- (ii) Determine the work done by the gas. (1)
- (iii) Hence calculate the change in internal energy of the gas. (2)

(Total 6 marks)

© International Baccalaureate Organisation

13 This question is about a heat engine.

A certain heat engine uses a fixed mass of an ideal gas as a working substance. The graph opposite shows the changes in pressure and volume of the gas during one cycle ABCA of operation of the engine.

- (a) For the part $A \rightarrow B$ of the cycle, explain whether
- (i) work is done **by** the gas or work is done **on** the gas. (1)
- (ii) thermal energy (heat) is absorbed **by** the gas or is ejected **from** the gas to the surrounding. (1)